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Improving source camera identification using a simplified total variation based noise removal algorithm

Floris Gisolf*, Anwar Malgoezar, Teun Baar, Zeno Geradts

Department of Digital Technology and Biometry, Netherlands Forensic Institute, Laan van Ypenburg 6, 2497 GB, The Hague, The Netherlands

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ABSTRACT

In this paper a new method for photo-response non-uniformity (PRNU) noise extraction is proposed. Photo-response non-uniformity noise patterns are a reliably method for digital camera identification. Especially with a large number of images the process of camera identification can be time consuming. The proposed method aims to increase the speed of PRNU extraction without losing accuracy when compared to the state-of-the-art method. Currently wavelet based denoising is used as the standard for PRNU extraction. Our proposed method is based on a simplified version of the Total Variation based noise removal algorithm. Results show that extraction is about 3.5 times faster with our method than with the wavelet based denoising algorithm. While initially only an increase in speed was the goal, results indicate that the Total Variation based noise removal algorithm is not only faster, but also more accurate than the state-of-the-art method.

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1. Introduction

With the increased use of digital media, digital camera identification becomes more important. Digital camera identification is the process of linking images to its source camera. This can be done in several ways, such as making use of photo-response non-uniformity (PRNU) noise (Lukáš et al., 2006), defect pixels (Geradts et al., 2001) and sensor dust (Dirik et al., 2008). This paper will focus on the use of PRNU noise to identify the source camera. PRNU noise is like a fingerprint for a camera and can make distinctions between cameras even if they are of the same brand and model (Lukáš et al., 2006) (Houten et al., 2011).

PRNU noise is caused by the variation in sensitivity of pixels to light. Imperfections during the manufacturing process of the image sensor cause pixels on the image sensor to give a slightly higher or lower signal than they are supposed to. This noise is approximately the same in every image that a camera takes and can therefore be used to identify the camera. The differences in sensitivity of the pixels are assumed to have a Gaussian distribution (Lukáš et al., 2006).

In this paper the following hypotheses are under investigation.

H_0 : The image comes from a different camera than the suspect camera.

H_1 : The image comes from the suspect camera.

To determine which hypothesis should be accepted, three steps have to be taken. The first step is the PRNU noise extraction from an image under investigation, usually done with a wavelet-based denoising algorithm (wavelet) (Lukáš et al., 2006). This is the noise filter. The second step is to obtain the Sensor Pattern Noise (SPN), which is the estimation of the PRNU pattern. The third step is to detect whether a suspect image correlates to a reference SPN. When a large number of images is involved, such as in a database, the process of camera identification can be time

* Corresponding author. Tel.: +31 618079733.
E-mail addresses: fgisolf@hotmail.com (F. Gisolf), a.malgoezar@gmail.com (A. Malgoezar), teunbaar@gmail.com (T. Baar), z.geradts@nfi.minvenj.nl (Z. Geradts).

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consuming. Examples of databases are social media sites such as Flickr, Facebook and Youtube, but also a child pornography database. We propose a new method of noise extraction for the purpose of PRNU estimation, an approximation based on the total variation. In order for this method to be effective it should be much faster than current state-of-the-art methods, without loss of accuracy in the detection of the correct camera.

Several attempts have been made to improve digital camera identification. Some were aimed at improving accuracy, others at improving speed. Chen et al. (2008) proposed zero-mean preprocessing to reduce effects of periodic artefacts, such as colour interpolation, that are not unique for a specific camera. In the same paper it was also suggested to use Wiener filtering after a Fourier Transform to reduce JPEG compression artefacts. In Gloe et al. (2012) it was shown that the Wiener filtering can also reduce other camera artefacts. Texture of scene content can contaminate the SPN. Therefore Li proposed 5 models to attenuate the influence of scene content (Li, 2010). To increase the speed of camera identification, as well as reduce SPN size, Bayram et al. proposed the use of binary-quantization of the SPN (Bayram et al., 2012). Each element of the SPN is a double precision number and is assumed to be normally distributed around 0. By only saving the sign information, positive or negative, each element is reduced to the size of 1 bit. This increases the speed of step 3, the calculation of the detection statistic, and also greatly reduces the size of the SPN. Another method used to decrease computation time and the size of the patterns of camera identification is applying a digest on the SPN, using only a percentage of the highest values of the SPN (Hu et al., 2009) (Goljan and Filler, 2010).

In Section 2 current state-of-the-art methods that will be used in this paper will be described in more detail. The rest of the paper is organized as follows. In Section 3 the proposed method will be explained. Section 4 will explain the experimental setup and Section 5 will discuss experiments and results. Section 6 is the conclusion of this paper.

2. Current methods

As stated in the introduction, the process of digital camera identification consists of three steps. Current state-of-the-art methods for each of the steps will be described in more detail here.

2.1. Step 1: PRNU noise extraction

The first step is to extract the noise residue, which is obtained as

$$\mathbf{w} = \mathbf{I} - F(\mathbf{I}), \tag{1}$$

where $\mathbf{I}$ is the original image, F the denoising operator and $F(\mathbf{I})$ the denoised image.

Our proposed method will be compared to three different algorithms, namely wavelet, Anisotropic Diffusion (AD) and Adaptive Spatial (AS) filtering. In recent papers on digital camera identification the wavelet has been used as denoising filter (Goljan et al., 2009) (Li, 2010) (Kang et al., 2012) (Gloe et al., 2012). The wavelet as used in (Lukáš et al., 2006) will be compared to the proposed algorithm.

Two other algorithms, Anisotropic Diffusion (AD) filtering (van Houten and Geradts, 2013) and Adaptive Spatial (AS) filtering (Cooper, 2013), will also be compared. The two other algorithms have been chosen because the authors stated that it would decrease computation time compared to the wavelet.

The AD filter is based on reducing image noise without removing significant parts of the image content which could contaminate the PRNU pattern. AD is defined as

$$\frac{\partial I}{\partial t} = \nabla \cdot (c(x,y,t)\nabla I), \tag{2}$$

where $c(x,y,t)$ is the diffusion coefficient and t signifies the multiscale approach.

The AS is based upon applying a combination of an iterative adaptive Wiener and median filtering in the spatial domain.

2.2. Step 2: SPN extraction and estimation

The second step is to obtain the SPN. It is assumed that the suspect camera is available to acquire flat-field images. These are images of a homogeneously coloured, evenly lit area, such as a grey wall or the blue sky. Flat-field images give the best approximation of the PRNU pattern on an image sensor. To obtain an estimation of the SPN these images have to be averaged. There are several ways to do this. In this paper three methods will be used: Basic SPN, MLE SPN and Phase SPN. Basic SPN is the most simplistic method and just averages the extracted noise of all the images as

$$\mathbf{y} = \frac{\sum_{i=1}^{L} \mathbf{w}_i}{L}, \tag{3}$$

where $\mathbf{y}$ is the SPN, and L the number of images used.

In 2008 Chen et al. proposed the use of a Maximum Likelihood Estimator SPN (MLE SPN). First, the PRNU *factor* $\widehat{\mathbf{K}}$ is determined by

$$\widehat{\mathbf{K}} = \frac{\sum_{i=1}^{L} \mathbf{w}_i \mathbf{I}_i}{\sum_{i=1}^{L} \mathbf{I}_i^2}. \tag{4}$$

Then the SPN can be calculated according

$$\mathbf{y} = \mathbf{I}_{nat}\widehat{\mathbf{K}} \tag{5}$$

where $\mathbf{I}_{nat}$ is the natural image that you want to compare with the SPN.

In 2012 Kang et al. proposed the use of Phase SPN, which seemed to outperform other methods. Phase SPN applies signal whitening to the noise residue before averaging it as

$$\mathbf{w}_{\phi i} = \frac{\mathcal{F}(\mathbf{w}_i)}{|\mathcal{F}(\mathbf{w}_i)|}, \tag{6}$$

where $\mathcal{F}$ is the Fourier transform. Then the SPN is obtained as

$$\mathbf{y} = real\left[\mathcal{F}^{-1}\left(\frac{\sum_{i=1}^{L}\mathbf{w}_{\phi i}}{L}\right)\right], \tag{7}$$

where the real part of the inverse Fourier transform is taken of the average of the obtained $\mathbf{w}_{\phi i}$. The imaginary part is very small and can be ignored.

2.3. Step 3: detection statistics

To decide which hypothesis (H_0 or H_1) to accept, a detection statistic must be calculated. For this the correlation or the peak-to-correlation energy (PCE) can be used. If we denote $\mathbf{x}$ as the obtained noise residue from a natural image $\mathbf{I}_{nat}$ and the SPN $\mathbf{y}$ as vectors, then the correlation is defined as

$$\rho(\mathbf{x},\mathbf{y}) = \frac{\sum_{i=1}^{n} x_i y_i}{\sqrt{\sum_{i=1}^{n} x_i^2}\sqrt{\sum_{i=1}^{n} y_i^2}}, \tag{8}$$

and the PCE as

$$\mathcal{P}(\mathbf{x},\mathbf{y}) = sgn[c(0)]\frac{c^2(0)}{\frac{1}{n-|\mathcal{A}|}\sum_{l,l\notin\mathcal{A}} c^2(l)}, \tag{9}$$

where n is the length of the vector and $|\mathcal{A}|$ is the cardinality of a square area where the peak occurs of the circular cross-correlation $c(l)$ between $\mathbf{x}$ and $\mathbf{y}$. The circular cross-correlation $c(l)$ is defined as

$$c(l) = \frac{1}{n}\sum_{i=1}^{n} x_i y_{i\oplus l}, \tag{10}$$

where sgn[⋯] picks out the sign of the correlation to lower the false positive rate by half.

Depending on which matching metric is used, H_0 will be accepted if $\rho<\tau_\rho$ or $\mathcal{P}<\tau_\mathcal{P}$, where τ_ρ and $\tau_\mathcal{P}$ are the threshold values. The threshold is set depending on the desired false positive rate during the experiments.

3. The proposed noise extraction algorithm

The proposed method uses a noise removal method for images which involves the minimization of the total variation (TV) (Rudin et al., 1992). Using the TV model, one minimizes the L^1 norm of the gradient instead of the L^2 norm used in least-square restoration models. The denoised image is obtained by minimizing the total variation norm of the estimated solution. The constraints are determined by the noise statistics. The TV model is used in image processing to denoise data without the blurring of jumps or edges.

For PRNU extraction a simplified version of the TV model will be used. Simplifications are applied in order to reduce the complexity of the problem and thus reducing the computation time. This should result in a faster camera identification process than when using the state-of-the-art wavelet.

Denoting the observed intensity as $u_0(x,y)$ for pixel values of a noisy image, the desired clean image is denoted as $u(x,y)$ for $x,y\in\Omega$, the image domain, as

$$u_0(x,y) = u(x,y) + n(x,y), \tag{11}$$

where n is white additive zero mean noise

$$\mathbb{E}[n(x,y)] = 0, \mathbb{E}\left[n^2(x,y)\right] = \sigma^2. \tag{12}$$

In light of the previous section this means $\mathbf{I}\rightarrow u_0$ and $F(\mathbf{I})\rightarrow u$. The constrained minimization problem is defined as (Rudin et al., 1992) to be

$$min\, TV[u] = min\int_\Omega |\nabla u| dxdy \tag{13}$$

subject to the constraints involving u_0

$$\int_\Omega u dxdy = \int_\Omega u_0 dxdy \tag{14}$$

and

$$\int_\Omega \frac{1}{2}(u-u_0)^2 dxdy = \sigma^2, \tag{15}$$

where ∇ is the gradient operator and σ is the standard deviation of the noise $n(x,y)$. The first constraint is automatically satisfied due to the translation invariance of the TV norm (Chambolee and Lions, 1997).

Fitting the second constraint and introducing the Lagrange multiplier λ, the additive noise removal problem can be formulated as an energy minimizing function of

$$J[u] = \int_\Omega |\nabla u| dxdy + \frac{\lambda}{2}\int_\Omega (u-u_0)^2 dxdy. \tag{16}$$

The corresponding Euler–Lagrange equation will then be

$$-\nabla\cdot\left(\frac{\nabla u}{|\nabla u|}\right) + \lambda(u-u_0) = 0 \tag{17}$$

and applying the gradient descent method results in

$$\frac{\partial u}{\partial t} = \nabla\cdot\left(\frac{\nabla u}{|\nabla u|}\right) - \lambda(u-u_0), \tag{18}$$

where t is the time evolution and numerically used as iteration steps. As t increases, we approach the denoised version of the image.

In the process, singularities are avoided by introducing a small positive parameter ε

$$\left|\nabla u\right|_\varepsilon = \sqrt{\left|\nabla u\right|^2 + \varepsilon^2}. \tag{19}$$

As a first approximation it is assumed that there is no fitting constraint, i.e. $\lambda = 0$, for a large reduction in numerical computation time. This results in the Osher-Rudin TV equation (Rudin and Osher, 1994)

$$\frac{\partial u}{\partial t} = \nabla\cdot\left(\frac{\nabla u}{|\nabla u|_\varepsilon}\right). \tag{20}$$

Each successive approximation of the denoised image can be formulated as an iterative equation

$$u_{n+1} = \nabla\cdot\left(\frac{\nabla u_n}{|\nabla u_n|_\varepsilon}\right)\Delta t + u_n, \tag{21}$$

where the Courant–Friedrichs–Lewy (CFL) condition must be imposed on Δt

$$\Delta t \le c|\nabla u|_\varepsilon \tag{22}$$

for some constant $c > 0$. Using Eq. (19), $\Delta t = c\varepsilon$ is used as the time evolution step. At each step the approximated noise residual can be found using $u_0 - u_n$.

The simplified algorithm will be named First Step Total Variation (FSTV) and is denoted as

$$w = u_0 - u_1 = -\nabla\cdot\left(\frac{\nabla u_0}{|\nabla u_0|_\varepsilon}\right), \tag{23}$$

where the time step Δt is left out. The time step is an added time evolution step and only of importance when using multiple steps to arrive at the solution.

FSTV is a relatively simple algorithm which removes only a small amount of noise, as will be demonstrated later. Because PRNU noise is the largest noise component, it is hypothesized that FSTV still removes a sufficient amount of PRNU noise for camera identification. Due to its simplicity computation time should be much less.

4. Experimental setup

For the experiment an MATLAB implementation provided by the Binghamton University (Camera Fingerprint – Matlab Implementation) has been used. The script uses a wavelet denoising algorithm for extraction, MLE SPN for the reference SPN and PCE as the detection statistic. Modifications have been made so that FSTV, AD, AS, Basic SPN and Phase SPN could be used as well. In the case of FSTV, finite differences was used to calculate the gradient. Cameras provided by the Netherlands Forensic Institute (NFI) have been used, as well as images from the Dresden Image Database (Gloe and Böhme, 2010). All the images are of 'good' quality, meaning that they have a resolution of at least 1 mega pixel and a JPEG quality factor of 88 or higher. For each model there were multiple cameras. For each camera a number of flat-field images have been used to create the reference SPNs. Per camera, for each method (wavelet and FSTV) a Basic SPN, an MLE SPN and a Phase SPN were created using the flat-field images. Per camera at least 65 natural images were made of various objects and landscapes. In Table 1 the brands and models can be found, as well as the resolution of the images, the number of flat-field images used for the reference SPN (L) and the number of natural images.

When the noise residue $\mathbf{w}$ of an image of a camera is compared with the reference SPN of that same camera it is called a match. If the noise residue $\mathbf{w}$ of an image of a camera is compared with the reference SPN of a different camera it is called a mismatch. It is assumed that comparisons between cameras of the same brand and model have a higher statistical mismatch value than comparisons between cameras from a different brand and model. This is caused by intermodel artefacts, such as the CFA and JPEG compression. Therefore images will only be compared to reference sets of the source camera and the other cameras of that same brand and model. This is to show the strength of the methods and it simulates a 'worst-case' scenario. Comparisons are done for the three separate colour channels, red, green and blue. The sum of the results of these three channels is then used for further calculations.

When high quality images are used, it is expected that all algorithms will have a 100% accuracy and thus will show no performance differences. Therefore the natural images have undergone compression and cropping. Each natural image has been cropped in the middle to a resolution of 512×512 and has been compressed to a JPEG quality factor of 50. For each method a new Basic SPN, MLE SPN and Phase SPN were created, this time using 512×512 crops of the middle of the flat-field images. This degraded image quality should make it possible to detect performance differences between the tested algorithms.

5. Experiments and results

The image of Lena was used to test the performance of FSTV. Gaussian noise with $\sigma = 10$ was added to the original Lena image. A fully functional Osher–Rudin TV model with 25 time steps was also used as a comparison, as well as the wavelet. Results of the denoising can be seen in Fig. 1 with corresponding root-mean-square error (RMSE) values. As expected for FSTV the difference before and after denoising is very small, compared to 25 steps of the TV model and wavelet.

To determine if the small amount of noise extracted by FSTV is enough for camera identification with the same accuracy as the wavelet the following experiment was done. The noise residue $\mathbf{w}_i$ of each original natural image was extracted using FSTV and then compared one by one to the Basic SPN of FSTV of the same camera and to the other cameras of the same brand and model. The same was done for the other algorithms (wavelet, AS, AD): the noise residue $\mathbf{w}_i$ of each original natural image was extracted using the respective algorithm and then compared one by one to the Basic SPN of that algorithm. As was expected for all cameras a true positive rate (TPR) of 100% was found at a false positive rate (FPR) of 0% for all methods.

From here on only the degraded natural images and reference SPNs are used. The noise residue $\mathbf{w}_i$ of each degraded natural image was extracted using FSTV and then compared one by one to the Basic SPN of FSTV of the same camera and to the other cameras of the same brand and model. The same was done for the other algorithms: the noise residue $\mathbf{w}_i$ of each degraded natural image was extracted and then compared one by one to the Basic SPN of the respective algorithm. For FSTV and wavelet the same was done with MLE SPN and Phase SPN. Receiver Operating Characteristic (ROC) curves were created with the obtained data. An ROC curve plots the fraction of true positives out of the matches against the fraction of false positives out of the mismatches, while varying the threshold. Overall ROC curves for all cameras can be seen in Fig. 2, sorted by SPN type.

For Basic SPN and MLE SPN the two algorithms perform almost equal. Only when using Phase SPN a difference can be seen. The combination of FSTV and Phase SPN increases

Table 1
Cameras used in experiments.

	Camera name	# of cameras	Resolution	# of flat-field images per camera (L)	# of naturals per camera
NFI cameras	Blackberry Curve 9360	4	2592 × 1944	100	65
	Canon IXUS 220HS	4	2816 × 2112	100	65
	Canon PowerShot A430	4	2272 × 1704	100	65
	Panasonic Lumix FP7	4	2560 × 1920	100	65
	Panasonic Lumix FZ45	4	2048 × 1536	100	65
	Samsung Digimax L70	4	3072 × 2304	100	65
	Samsung Digimax S500	4	2560 × 1920	100	65
	Sony Cyber-shot DSC-S930	4	3648 × 2736	100	65
	Sony Cyber-shot DSC-S500	4	2816 × 2112	100	65
	Samsung ST30	4	3648 × 2736	100	65
Dresden Image Database	Fujifilm FinePix J50	3	3264 × 2448	50	65
	Olympus Mju 1050 SW	4	3648 × 2736	50	65
	Pentax Optio A40	4	4000 × 3000	50	65
	Ricoh Caplio GX100	4	3648 × 2736	25	65
	Nikon Coolpix S710	4	4352 × 3264	50	65

overall performance of the identification compared to all other combinations.

Fig. 3 shows the ROC curves of FSVT compared to AD and AS. FSTV slightly outperfoms both alternatives.

Due to the large number of cameras used, individual camera model ROC curves are not shown. Some data on individual cameras from wavelet and FSTV can be seen in Table 2. This table shows the TPR of each camera model, split by filter and SPN method at two different thresholds. In forensics a low FPR is required, so thresholds were set at an FPR of 0.01 and 0.05.

FSTV is only the first step of the TV model. Increasing the number of steps will increase the calculation time linearly. When using multiple steps, two parameters have to be set: the Δt and the number of steps. Increasing Δt will increase the strength of the noise removal. When Δt is set too high, scene content will also be removed, thus contaminating the **w**. Experiments indicated that a Δt of 1 gave the best results. Increasing the number of steps has the same type of effect as increasing Δt:more noise will be removed, but scene content may also be affected. Using two steps results in a doubling of calculation time

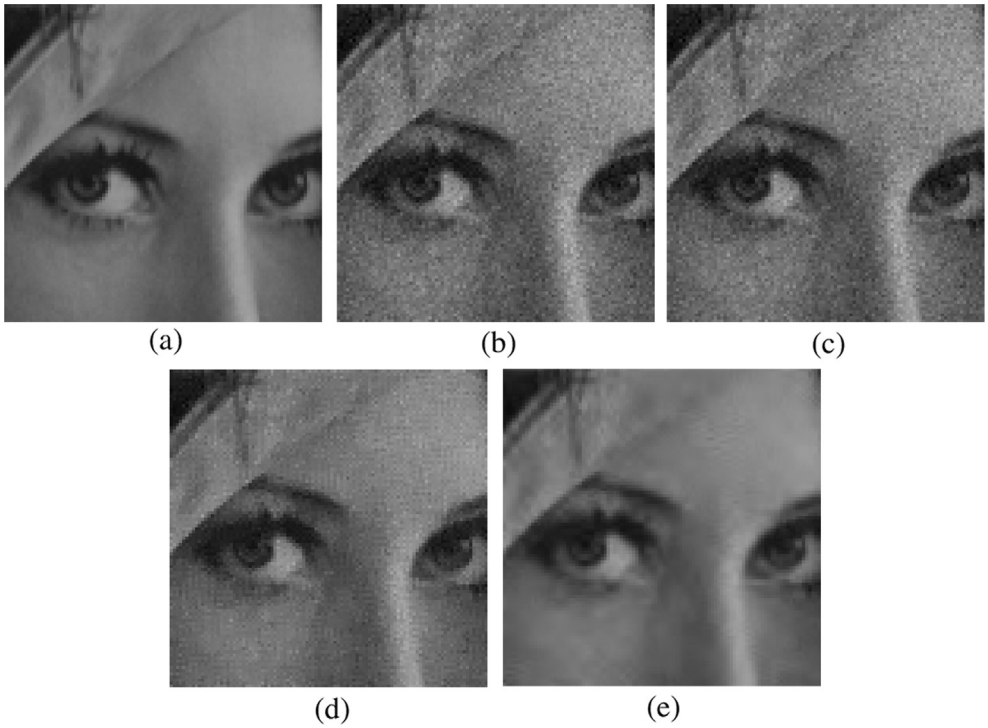


Fig. 1. (a) Original Lena image. (b) Lena with Gaussian noise added ($\sigma = 10$), RMSE = 10.1. (c) FSTV denoising, RMSE = 9.94. (d) 25 steps performed of TV denoising, RMSE = 7.20. (e) Wavelet denoised image, RMSE = 4.74.

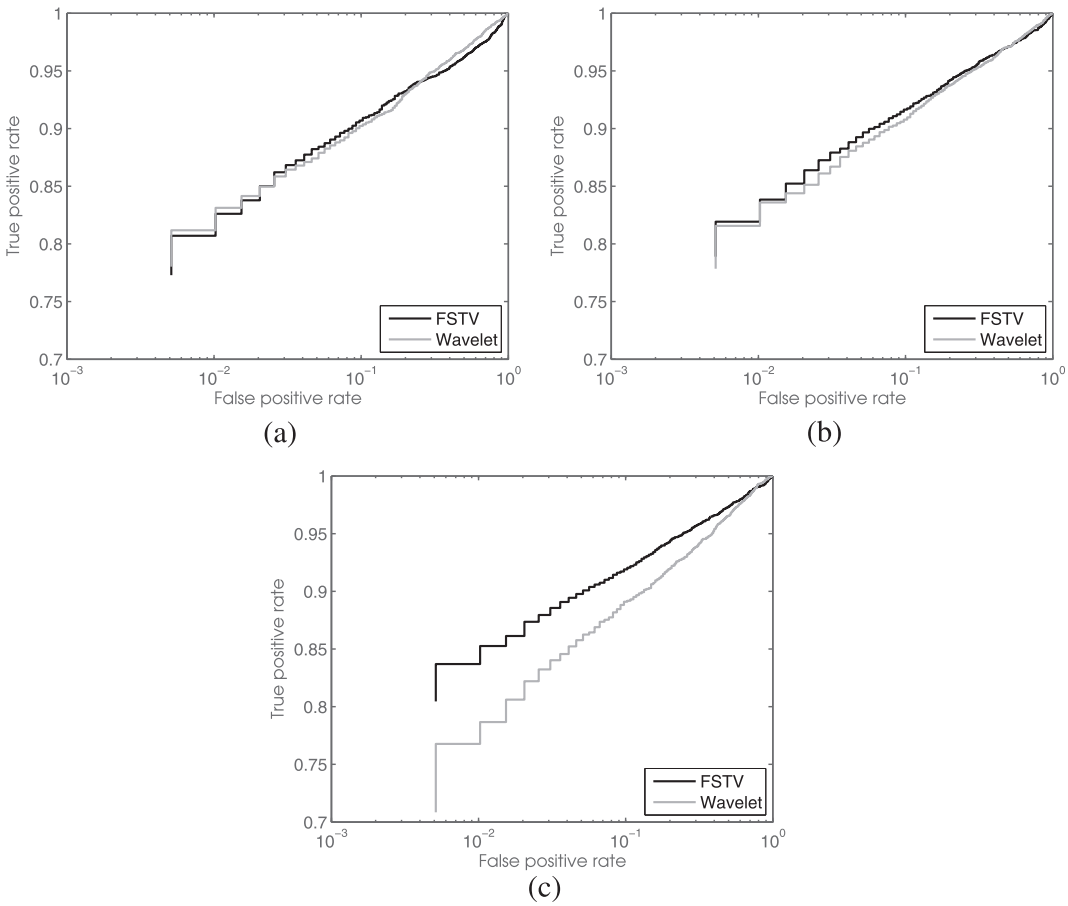


Fig. 2. ROC curve of FSTV versus wavelet using (a) Basic SPN, (b) MLE SPN and (c) Phase SPN.

compared to FSTV, but is still faster than Wavelet. Using multiple steps of the TV model could potentially extract a better noise residue. Two steps of the TV model with $\Delta t = 1$ will therefore be compared to FSTV. The methods were the same as for the other ROC curves and only Basic SPN was used. Fig. 4 shows the ROC curves. Two steps of the TV model results in an increased TPR of 9% compared to FSTV.

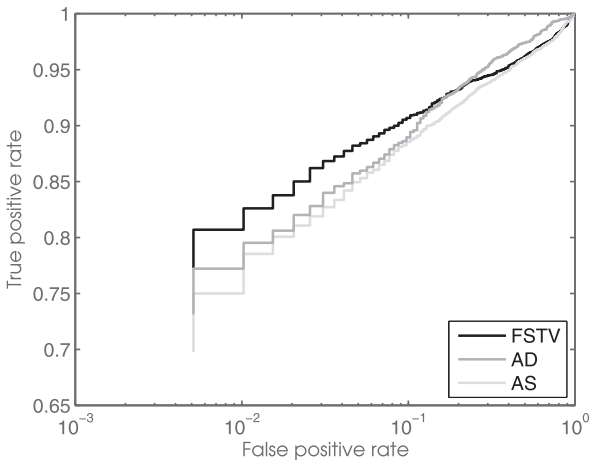


Fig. 3. FSTV versus AS and AD using Basic SPN.

In Table 3 results can be found of several tests that have been done to show the difference in computation time between the FSTV, the TV model with multiple steps, wavelet, AD and AS. Sets of 100 images of varying resolutions have been used to create a Basic SPN. The time it took to create the SPN is shown in seconds. MATLAB was used to show computation times of wavelet, FSTV and multiple steps of TV model, AD and AS. Part of the wavelet implementation for MATLAB was compiled in C to reduce calculation time. For a fair comparison these parts have been rewritten in MATLAB code for the computation time comparison. NFI PRNU Compare Law Enforcement Edition was also used to show computation times of wavelet and FSTV. NFI PRNU Compare is written in Java and is therefore faster than MATLAB. The tests were done on an HP Elitebook 8560w with an Intel Core i7-2630QM (2.0 GHz quad core) CPU and 4 GB RAM.

As can be seen FSTV is considerably faster than any of the other algorithms. Depending on the resolution of the images FSTV is 3.6–5.8 times faster than wavelet when using MATLAB. In NFI PRNU Compare FSTV is 3.5–3.8 times faster than wavelet, and 10 times faster than FSTV in MATLAB.

Matsushita and Kitazawa (2009) state that wavelet smoothes around edges. This can be seen in the noise

Table 2
TPR for all cameras with different methods at FPR = 0.01 and FPR = 0.05. The best result per camera is shown in bold.

Camera	SPN method	0.01		0.05		Camera	SPN method	0.01		0.05	
		Wavelet	**FSTV**	**Wavelet**	**FSTV**			**Wavelet**	**FSTV**	**Wavelet**	**FSTV**
BlackBerry Curve 9360						*Pentax A40*					
	Basic	**0.996**	**0.996**	0.996	**1.000**		Basic	0.885	0.905	0.930	**0.957**
	MLE	0.985	0.992	0.996	0.996		MLE	0.908	0.903	0.935	**0.957**
	Phase	0.992	**0.996**	0.992	0.996		Phase	0.805	**0.922**	0.883	0.955
Canon IXUS 220HS						*Ricoh GX100*					
	Basic	**0.988**	0.969	**0.988**	0.985		Basic	0.970	0.982	0.983	**0.997**
	MLE	0.981	0.962	0.985	0.985		MLE	0.953	**0.985**	0.982	0.995
	Phase	0.981	0.981	**0.988**	**0.988**		Phase	0.860	0.982	0.950	0.993
Canon PowerShot A430						*Samsung Digimax S70*					
	Basic	0.985	**0.988**	**0.988**	**0.988**		Basic	0.946	0.954	0.962	0.973
	MLE	0.981	0.981	0.985	0.985		MLE	0.912	0.935	0.954	**0.977**
	Phase	0.981	**0.988**	**0.988**	**0.988**		Phase	0.865	**0.958**	0.915	0.973
Coolpix S710						*Samsung Digimax S500*					
	Basic	0.631	0.719	0.838	**0.862**		Basic	0.681	0.715	0.708	0.881
	MLE	0.588	0.700	0.754	0.788		MLE	**0.912**	0.677	**0.938**	0.846
	Phase	0.558	**0.742**	0.735	0.842		Phase	0.804	0.796	0.923	0.900
Fujifilm J50						*Samsung ST30*					
	Basic	0.713	**0.733**	0.742	0.750		Basic	0.992	**0.996**	**0.996**	**0.996**
	MLE	0.710	0.708	0.735	**0.753**		MLE	0.988	0.992	**0.996**	**0.996**
	Phase	0.668	0.727	0.732	0.750		Phase	0.992	**0.996**	**0.996**	**0.996**
Olympus						*Sony Cyber-Shot DSC S500*					
	Basic	0.635	**0.653**	0.755	0.744		Basic	0.312	**0.427**	0.585	0.627
	MLE	0.609	0.604	0.716	0.716		MLE	0.342	0.362	0.612	0.608
	Phase	0.560	0.674	0.721	**0.764**		Phase	0.038	0.388	0.262	**0.658**
Panasonic LUMIX FP7						*Sony Cyber-Shot DSC S930*					
	Basic	0.985	0.958	0.992	0.977		Basic	0.642	0.558	0.796	0.715
	MLE	**1.000**	0.962	**1.000**	0.977		MLE	0.646	0.623	**0.831**	0.742
	Phase	0.985	0.985	0.996	0.985		Phase	0.627	**0.654**	0.758	0.773
Panasonic LUMIX FZ45											
	Basic	0.742	**0.965**	0.912	**0.992**						
	MLE	0.851	0.950	0.923	0.988						
	Phase	0.958	0.962	**0.992**	0.981						

residue of natural images and could explain why FSTV outperforms wavelet. While FSTV also shows some artefacts around edges, these are only 1 pixel wide, whereas the wavelet smoothing affects several pixels. A clear example of this can be seen in Fig. 5. While the noise residue obtained with FSTV looks like AWGN, wavelet shows considerable artefacts in the area of the tree. When comparing the noise residue to the SPN of the source camera, the FSTV PCE score is nearly 1.5 times higher.

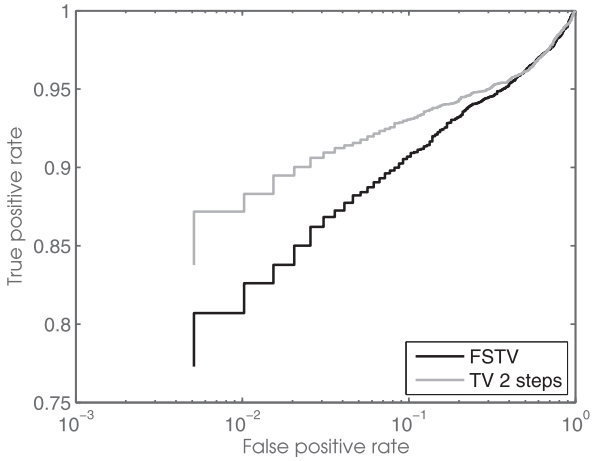


Fig. 4. FSTV versus TV 2 steps.

6. Conclusion

In this paper a new filter, FSTV, is proposed for digital camera identification. Although the primary aim was to develop a faster algorithm results show that FSTV not only takes less computation time than the wavelet, but is also more accurate. The combination of FSTV and Phase SPN gave the best results. Using FSTV proves to be a major reduction in calculation time, which is very useful when

Table 3
CPU time for various denoising methods.

Resolution	Software	Denoise method	CPU time (s)
1024 × 768	MATLAB	FSTV	41.08
		TV 2 steps	74.94
		AS	80.94
		Anisotropic	104.26
		TV 5 steps	176.22
		Wavelet	239.61
		TV 10 steps	345.27
	PRNU Compare	FSTV	3.69
		Wavelet	13.20
3648 × 2736	MATLAB	FSTV	519.08
		TV 2 steps	945.58
		AS	987.66
		Anisotropic	1250.33
		Wavelet	1893.97
		TV 5 steps	2221.60
		TV 10 steps	4348.98
	PRNU Compare	FSTV	48.56
		Wavelet	183.41

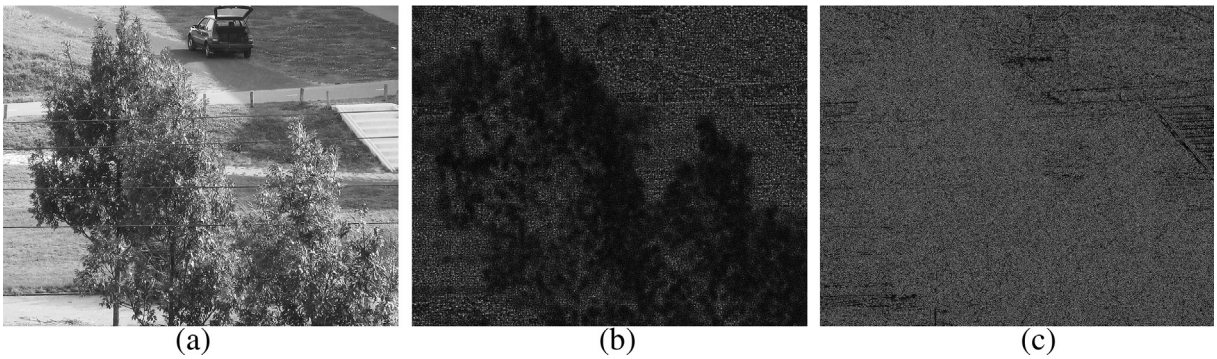


Fig. 5. Differences in noise artefacts with (a) the original image, (b) the wavelet noise residue and (c) the FSTV noise residue.

dealing with large databases. Another advantage is that there are no parameters that need to be set for FSTV, unlike the other algorithms, which makes it easy to use. Using two steps of the TV model can also be considered; it has a higher accuracy and is still faster than the other tested algorithms. The downside is that Δt has to be defined, introducing a parameter. More research with FSTV has to be done to verify our results with more cameras and in different circumstances. However, our results suggest that FSTV should be considered as the new standard in digital camera identification, especially when time is of essence.

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